BIOINFORMATICS PORTFOLIO · 2026

Byung-cheol Kang

Genomics · Bacteriophages · Research Pipelines

I am a bioinformatics researcher and developer who analyzes genomic and bacteriophage data and builds research pipelines and databases. I use AI coding tools to implement analysis pipelines, develop phage analysis and database tools, and automate repetitive research tasks.

CURRENT AND RECENT ROLES

MicrobiotiX · Bioinformatician 2026.07 - PRESENT

- Bacterial genome analysis for host context and phage genome analysis across QC, annotation, taxonomy, lifestyle, packaging, and tail biology.
- **K. pneumoniae** tail/RBP and KL-type database construction and evaluation.
- Protein structure prediction, structure-based annotation, and reference-structure comparison.
- In silico tail-RBP design using explicit attachment, target, taxonomy, and structural criteria.

Wittgenbio · Computational Biology Research 2026.07.01 - 2026.10.01

- Developed bulk-to-single-cell inference approaches that connect bulk expression profiles with curated single-cell references.
- Contributed to virtual-cell research through computational cell-state representation, model evaluation, and biologically grounded interpretation.

CAREER SNAPSHOT

6+ years	5-person	4 profiles	7 papers
Applied bioinformatics	Team led at Sanigen	16S · ITS · WGS · Hybrid	SCI(E) publications

Past platform evidence. The integrated workflow and operational results in this portfolio describe work performed at Sanigen (2020.03 - 2025.10). They are not presented as MicrobiotiX or Wittgenbio assets or current-company outcomes.

PROJECT 01 · PLATFORM

Integrated Microbial Genomics Platform

PROBLEM

Separate assay scripts, environments, quality checks, server paths, and reporting conventions made analysis difficult to reproduce and review consistently.

SYSTEM DESIGN

Entry point	One Snakemake dispatcher selects 16S, ITS, WGS, or Hybrid profiles.
Execution	Server-specific YAML configuration with Conda, Docker, or Apptainer and pinned BioContainers.
Quality	Input validation, raw-read QC, assembly/analysis checks, review gates, and retained logs.
Delivery	Custom Python/R processing, visualizations, HTML reports, and packaged outputs.

ACTIVE TOOL CHAINS

Profile	Core processing	QC / interpretation
16S / ITS	QIIME 2, DADA2, SILVA classifier, MAFFT, FastTree	FastQC, MultiQC, diversity metrics, custom R/Python reports
WGS	Trimmomatic, SPAdes, Bowtie2, samtools, Prokka	QUAST, BUSCO, Kraken2, Abricate/ CARD/VFDB, eggNOG-mapper
Hybrid	NanoFilt, NanoPlot, Flye or Unicycler, BWA, Pilon, minimap2	QUAST, BUSCO, Prokka, CGView, BLASTn, Abricate, eggNOG-mapper

EVIDENCE BOUNDARY

Only tools connected to active Snakemake includes and rules are listed as active. Configured or documented but inactive components are not represented as production use.

PROJECT 02 · OPERATIONS

Analysis Operations and Quality Assurance

OPERATING MODEL

Question
biological scope

→

Execution
workflow + QC

→

Delivery
interpretation + provenance

RESPONSIBILITIES AT SANIGEN

- Translated research and client questions into analysis scope, metadata requirements, and appropriate workflow profiles.
- Reviewed Q20/Q30, read depth, contiguity, assembly plausibility, taxonomy, annotation, and analysis-specific quality evidence before interpretation.
- Coordinated reruns, technical communication, report packaging, manuals, and training across a five-person team.
- Maintained reproducible environments and operational failure handling across Linux compute and storage infrastructure.

VERIFIED LOWER BOUND OF ARCHIVED WORK

28
WGS order codes

21
Hybrid order codes

19
16S order codes

18
Shotgun order codes

7
RNA order codes

3
ITS order codes

2
Other order codes

These 98 distinct order codes are a conservative lower bound derived from archived communications. They are not sample counts, revenue, or an independently audited total workload.

PROJECTS 03-04 · COMPARATIVE GENOMICS AND PANELS

From Genome Differences to Assay Candidates

COMPARATIVE-GENOMICS MARKER DISCOVERY

Objective. Identify a region that distinguishes a target group while remaining practical for downstream marker or assay development.

Approach. Curate complete genomes, compare whole-genome and pan-genome context, narrow candidate intervals, and review conservation, specificity, and downstream constraints.

Outcome. Prioritized an approximately **17 kb candidate region** for follow-up, with traceable reasoning from genome selection to practical assay constraints.

FOODBORNE-PATHOGEN NGS PANEL

- Contributed bioinformatics and comparative-genomics analysis to marker-oriented panel development and evaluation.
- Supported interpretation of NGS outputs across food and environmental sample contexts.
- Collaborative outcomes included two peer-reviewed papers in 2023 and one joint patent contribution in 2022.

BOUNDARIES

> Computational prioritization does not establish diagnostic sensitivity or specificity. Publication and patent records document collaborative outcomes, not sole authorship or universal performance across unevaluated matrices.

SELECTED EVIDENCE

- **Frontiers in Microbiology** 14 (2023): agricultural wastewater pathogen panel.
- **Journal of Microbiology and Biotechnology** 33(1) (2023): fermented-food pathogen panel.
- Joint patent contribution (2022): NGS-based primer set and foodborne-bacteria detection method.

PROJECT 05 · CURRENT ROLE CASE STUDY

Klebsiella Phage Genomics and Tail-RBP Design

GENOME EVIDENCE WORKFLOW

Host + phage genomes
QC + annotation

→

Independent evidence
taxonomy + lifestyle + termini

→

Tail interpretation
KL target + structure

CONTRIBUTION AND EVIDENCE

- Integrated CheckV, Pharokka, Phold, BACPHLIP, PhaTYP, PhaGCN, and PhageTerm outputs while preserving the evidence behind each conclusion.
- Built a 7,759-sequence tail/RBP and KL-type reference resource with source traceability, confidence labels, and experimentally verified targets explicitly flagged.
- Cross-validated candidate depolymerases with DposFinder, SpikeHunter, DepoScope, sequence search, domain boundaries, and protein-structure evidence.
- Used ProstT5 and Foldseek with experimental PDB references to identify structurally conserved proteins at sequence identities below ordinary search sensitivity.

ANONYMIZED CASE-STUDY RESULTS

- Four genomes converged on lytic **Przondovirus** classifications and short DTR packaging with 180-183 bp terminal repeats.
- Three capsule targets matched experimentally supported references at 95.4-98.8% sequence identity; unsupported cases remained unresolved rather than receiving forced labels.
- One design candidate passed genus, attachment, target-label, experimental-structure, and predicted-structure criteria; its 129-aa anchor alignment spanned 99.2% of the region.

DESIGN OUTPUT AND BOUNDARY

The design bench supports domain swaps, full-module replacements, and tandem additions, with genome FASTA, module multi-FASTA, GenBank, and protein FASTA export. Host K-locus assignments, predicted RBP folding, adsorption, host range, and lytic activity still require bacterial-genome and wet-lab validation.

PROJECT 06 · WITTGENBIO RESEARCH

Bulk-to-Single-Cell and Virtual-Cell Research

BULK-TO-SINGLE-CELL WORKFLOW

Bulk profiles
QC + harmonization

→

Single-cell reference
curation + alignment

→

Inference
composition + state

ANALYTICAL RESPONSIBILITIES

- Curated reference atlases for tissue, condition, cell annotation, and population coverage.
- Aligned bulk and single-cell feature spaces while tracking missing, duplicated, and weakly represented signals.
- Inferred cell-type composition and reference-supported cell-state signals from bulk expression profiles.
- Reviewed sensitivity to reference choice, feature coverage, batch effects, and biological mismatch before interpretation.

VIRTUAL-CELL WORKFLOW

Cellular data
question + scope

→

Cell model
representation + fitting

→

Evaluation
state + response + biology

EVALUATION PRINCIPLES

- Separate model fitting from evaluation data across relevant cell states, conditions, and perturbations.
- Evaluate predicted states or responses against measurable biological evidence, not latent-space appearance alone.
- Report underrepresented tissues, cell populations, perturbations, and experimental conditions.

PUBLIC EVIDENCE BOUNDARY

Bulk-to-single-cell outputs are model-based estimates, not direct single-cell measurements. Virtual-cell predictions remain research hypotheses until externally or experimentally validated. Private data, implementation details, and performance metrics are not disclosed here.

PROJECT 07 · INDEPENDENT R&D PROTOTYPE

Phage Lifestyle Evidence Prototype

RESEARCH QUESTION

Can existing tools be combined into an end-to-end system that distinguishes lytic and lysogenic phages while preserving conflicting evidence and uncertainty?

EVIDENCE DESIGN

Quality context	CheckV completeness and contamination context
Annotation	Pharokka and Phold; integrase, repressor, excision, and lysis evidence
Classifiers	BACPHLIP and PhaTYP as independent prediction evidence
Decision layer	Agreement/confidence table with unresolved cases routed to manual review

HOST-RANGE EXTENSION

A related hypothesis-ranking layer considers LPS and outer-membrane gene variation together with phage receptor and annotation evidence. It prioritizes candidates for experiment; it does not prove adsorption, infection, or lytic activity.

PUBLICATION-GRADE NEXT STEPS

- External, leakage-controlled benchmark with known lifestyle labels.
- Comparison against individual tools and transparent simple baselines.
- Sensitivity, specificity, MCC, calibration, and quality-stratified performance.
- Ablation analysis for gene markers, classifiers, and quality context.
- Experimental validation for host-range candidate claims.

CONTACT

sentim2@gmail.com (+82) 10-5366-3411 bruce-bc.github.io